

习题 2.2

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- (3)

它的偶数项组成的子列为

$$\frac{2^n + 3^n}{3^{n+1} - 2^{n+1}} = \frac{\frac{1}{3} \left(\frac{2}{3}\right)^n + \frac{1}{3}}{1 - \left(\frac{2}{3}\right)^{n+1}}$$

我们有

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{1}{3} \left(\frac{2}{3}\right)^n + \frac{1}{3} &= \frac{1}{3} \\ \lim_{n \rightarrow \infty} 1 - \left(\frac{2}{3}\right)^{n+1} &= 1\end{aligned}$$

利用定理 2.7（四则运算）可知

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{3} \left(\frac{2}{3}\right)^n + \frac{1}{3}}{1 - \left(\frac{2}{3}\right)^{n+1}} = \frac{1}{3}$$

它的奇数项组成的子列为

$$\frac{3^n - 2^n}{2^{n+1} + 3^{n+1}} = \frac{\frac{1}{3} - \frac{1}{3} \left(\frac{2}{3}\right)^n}{\left(\frac{2}{3}\right)^{n+1} + 1}$$

我们有

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{1}{3} - \frac{1}{3} \left(\frac{2}{3}\right)^n &= \frac{1}{3} \\ \lim_{n \rightarrow \infty} \left(\frac{2}{3}\right)^{n+1} + 1 &= 1\end{aligned}$$

利用定理 2.7（四则运算）可知

$$\frac{\frac{1}{3} - \frac{1}{3} \left(\frac{2}{3}\right)^n}{\left(\frac{2}{3}\right)^{n+1} + 1} = \frac{1}{3}$$

综上，数列满足偶数项极限与奇数项极限相等，利用第一节例 8 可得，极限为 $\frac{1}{3}$ 。

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- (1)

关键点

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

- (2)

我们有

$$\begin{aligned}\sqrt{2}\sqrt[4]{2}\sqrt[8]{2}\cdots\sqrt[2^n]{2} &= 2^{\frac{1}{2}}2^{\frac{1}{4}}2^{\frac{1}{8}}\cdots2^{\frac{1}{2^n}} \\ &= 2^{1-\frac{1}{2^n}} \\ &= \frac{2}{\sqrt[2^n]{2}}\end{aligned}$$

又因为

$$\lim_{n \rightarrow \infty} \sqrt[2^n]{2} = 1$$

（证明与第一节例 5 相同）利用极限的四则运算（定理 2.7），我们有

$$\lim_{n \rightarrow \infty} \sqrt{2}\sqrt[4]{2}\sqrt[8]{2}\cdots\sqrt[2^n]{2} = 2$$

- (3)

这道题要找到求和公式。可以观察出分子是等差数列，分母是等比数列，这种大部分都是采用错位相减的方式。

$$\begin{aligned}S_n &= \frac{1}{2} + \frac{3}{2^2} + \cdots + \frac{2n-1}{2^n} \\ \frac{1}{2}S_n &= \frac{1}{2^2} + \frac{3}{2^3} + \cdots + \frac{2n-1}{2^{n+1}}\end{aligned}$$

于是，两式相减

$$\begin{aligned}\frac{1}{2}S_n &= \frac{1}{2} + \frac{2}{2^2} + \frac{2}{2^3} + \cdots + \frac{2}{2^n} - \frac{2n-1}{2^{n+1}} \\ &= \frac{1}{2} + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}} - \frac{2n-1}{2^{n+1}}\end{aligned}$$

其中，

$$\frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}}$$

是等比数列，其和为 $1 - \frac{1}{2^{n-1}}$ 。于是，我们有

$$\begin{aligned}\frac{1}{2}S_n &= \frac{1}{2} + 1 - \frac{1}{2^{n-1}} - \frac{2n-1}{2^{n+1}} \\ S_n &= 3 - \frac{1}{2^{n-2}} - \frac{2n-1}{2^n}\end{aligned}$$

所以

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{1}{2} + \frac{3}{2^2} + \cdots + \frac{2n-1}{2^n} &= \lim_{n \rightarrow \infty} 3 - \frac{1}{2^{n-2}} - \frac{2n-1}{2^n} \\ &= 3\end{aligned}$$

• (4)

我们有

$$\begin{aligned}\sqrt[n]{1 - \frac{1}{n}} &= \sqrt[n]{\frac{n-1}{n}} \\ &= \frac{\sqrt[n]{n-1}}{\sqrt[n]{n}}\end{aligned}$$

当 $n \geq 2$ 时，我们有

$$1 \leq \sqrt[n]{n-1} \leq \sqrt[n]{n}$$

利用迫敛性和例 2 可得

$$\lim_{n \rightarrow \infty} \sqrt[n]{n-1} = 1$$

利用极限的四则运算，我们有

$$\begin{aligned}\lim_{n \rightarrow \infty} \sqrt[n]{1 - \frac{1}{n}} &= \frac{\lim_{n \rightarrow \infty} \sqrt[n]{n-1}}{\lim_{n \rightarrow \infty} \sqrt[n]{n}} \\ &= 1\end{aligned}$$

• (5)

我们有

$$\begin{aligned}\frac{1}{n^2} + \frac{1}{(n+1)^2} + \cdots + \frac{1}{(2n)^2} &< \frac{1}{n^2} + \frac{1}{n^2} + \cdots + \frac{1}{n^2} \\ &= \frac{n+1}{n^2}\end{aligned}$$

又因为

$$\begin{aligned}0 &< \frac{1}{n^2} + \frac{1}{(n+1)^2} + \cdots + \frac{1}{(2n)^2} \\ \lim_{n \rightarrow \infty} \frac{n+1}{n^2} &= 0\end{aligned}$$

由迫敛性可得

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n^2} + \frac{1}{(n+1)^2} + \cdots + \frac{1}{(2n)^2} \right) = 0$$

• (6)

我们有

$$\begin{aligned}\frac{n}{\sqrt{n^2+n}} &\leq \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \cdots + \frac{1}{\sqrt{n^2+n}} \\ \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \cdots + \frac{1}{\sqrt{n^2+n}} &\leq \frac{n}{\sqrt{n^2+1}}\end{aligned}$$

又因为

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{n}{\sqrt{n^2+n}} &= \lim_{n \rightarrow \infty} \sqrt{\frac{n^2}{n^2+n}} \\ &= \lim_{n \rightarrow \infty} \sqrt{\frac{1}{1+\frac{1}{n}}} \\ &= 1\end{aligned}$$

同理

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt{n^2+1}} = 1$$

由夹逼定理可得

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \cdots + \frac{1}{\sqrt{n^2+n}} = 1$$

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$$\bullet (1) \lim_{n \rightarrow \infty} \frac{1}{2} \cdot \frac{3}{4} \cdot \dots \cdot \frac{2n-1}{2n}$$

因为 $n^2 \geq n^2 - 1 = (n-1)(n+1)$, 于是我们有

$$\begin{aligned} & \frac{1}{2} \cdot \frac{3}{4} \cdot \dots \cdot \frac{2n-1}{2n} \\ & \leq \frac{1}{\sqrt{3}} \cdot \frac{3}{\sqrt{3}\sqrt{5}} \cdot \dots \cdot \frac{2n-1}{\sqrt{2n-1}\sqrt{2n+1}} \\ & = \frac{1}{\sqrt{2n+1}} \end{aligned}$$

又因为

$$\begin{aligned} 0 & \leq \frac{1}{2} \cdot \frac{3}{4} \cdot \dots \cdot \frac{2n-1}{2n} \\ & \lim_{n \rightarrow \infty} \frac{1}{\sqrt{2n+1}} = 0 \end{aligned}$$

由夹逼定理可得

$$\lim_{n \rightarrow \infty} \frac{1}{2} \cdot \frac{3}{4} \cdot \dots \cdot \frac{2n-1}{2n} = 0$$

$$\bullet (2)$$

先考虑 $\sum_{p=1}^n p!$, 增大部分项的值 (前 $n-2$ 项), 我们有

$$\begin{aligned} n! & \leq \sum_{p=1}^n p! \leq (n-2)(n-2)! + (n-1)! + n! \\ & = (n-1)(n-2)! - (n-2)! + (n-1)! + n! \\ & = (n-1)! - (n-2)! + (n-1)! + n! \\ & \leq 2(n-1)! + n! \end{aligned}$$

因为

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{n!}{n!} = 1 \\ & \lim_{n \rightarrow \infty} \frac{2(n-1)! + n!}{n!} = 1 \end{aligned}$$

由迫敛性可得

$$\lim_{n \rightarrow \infty} \frac{\sum_{p=1}^n p!}{n!} = 1$$

- (3)

我们有

$$(n+1)^\alpha - n^\alpha = n^\alpha \left[\left(1 + \frac{1}{n}\right)^\alpha - 1 \right]$$

因为 $1 + \frac{1}{n} > 1$, 且 $0 \leq \alpha < 1$, 所以

$$\left(1 + \frac{1}{n}\right)^\alpha < 1 + \frac{1}{n}$$

所以

$$\begin{aligned} (n+1)^\alpha - n^\alpha &= n^\alpha \left[\left(1 + \frac{1}{n}\right)^\alpha - 1 \right] \\ &\leq n^\alpha \left[1 + \frac{1}{n} - 1 \right] \\ &= n^\alpha \frac{1}{n} \\ &= n^{\alpha-1} \\ &= \frac{1}{n^{1-\alpha}} \end{aligned}$$

又因为

$$\begin{aligned} 0 &< (n+1)^\alpha - n^\alpha \\ \lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} &= 0 \end{aligned}$$

由迫敛性可知

$$\lim_{n \rightarrow \infty} (n+1)^\alpha - n^\alpha = 0$$

- (4)

我们有

$$\begin{aligned} &(1+\alpha)(1+\alpha^2) \cdots (1+\alpha^{2^n}) \\ &= \frac{(1-\alpha)(1+\alpha)(1+\alpha^2) \cdots (1+\alpha^{2^n})}{1-\alpha} \\ &= \frac{1-\alpha^{2^{n+1}}}{1-\alpha} \\ &= \frac{1}{1-\alpha} - \frac{\alpha^{2^{n+1}}}{1-\alpha} \end{aligned}$$

因为 $|\alpha| < 1$, 于是

$$0 \leq \alpha^{2^{n+1}} < \alpha^n$$

又因为

$$\lim_{n \rightarrow \infty} \alpha^n = 0$$

由迫敛性可知

$$\lim_{n \rightarrow \infty} \alpha^{2^{n+1}} = 0$$

利用极限的四则运算可得

$$\begin{aligned} \lim_{n \rightarrow \infty} (1 + \alpha)(1 + \alpha^2) \cdots (1 + \alpha^{2^n}) &= \frac{1}{1 - \alpha} - 0 \\ &= \frac{1}{1 - \alpha} \end{aligned}$$

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设

$$a = \max\{a_1, a_2, \cdots, a_m\}$$

我们有

$$\sqrt[n]{a^n} = a \leq \sqrt[n]{a_1^n + a_2^n + \cdots + a_m^n} \leq \sqrt[n]{a^n + a^n + \cdots + a^n} = \sqrt[n]{ma^n} = \sqrt[n]{m}a$$

又因为

$$\begin{aligned} \lim_{n \rightarrow \infty} \sqrt[n]{m}a &= \lim_{n \rightarrow \infty} \sqrt[n]{m} \lim_{n \rightarrow \infty} a \\ &= 1 \cdot a = a \end{aligned}$$

由迫敛性可得

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_1^n + a_2^n + \cdots + a_m^n} = a$$